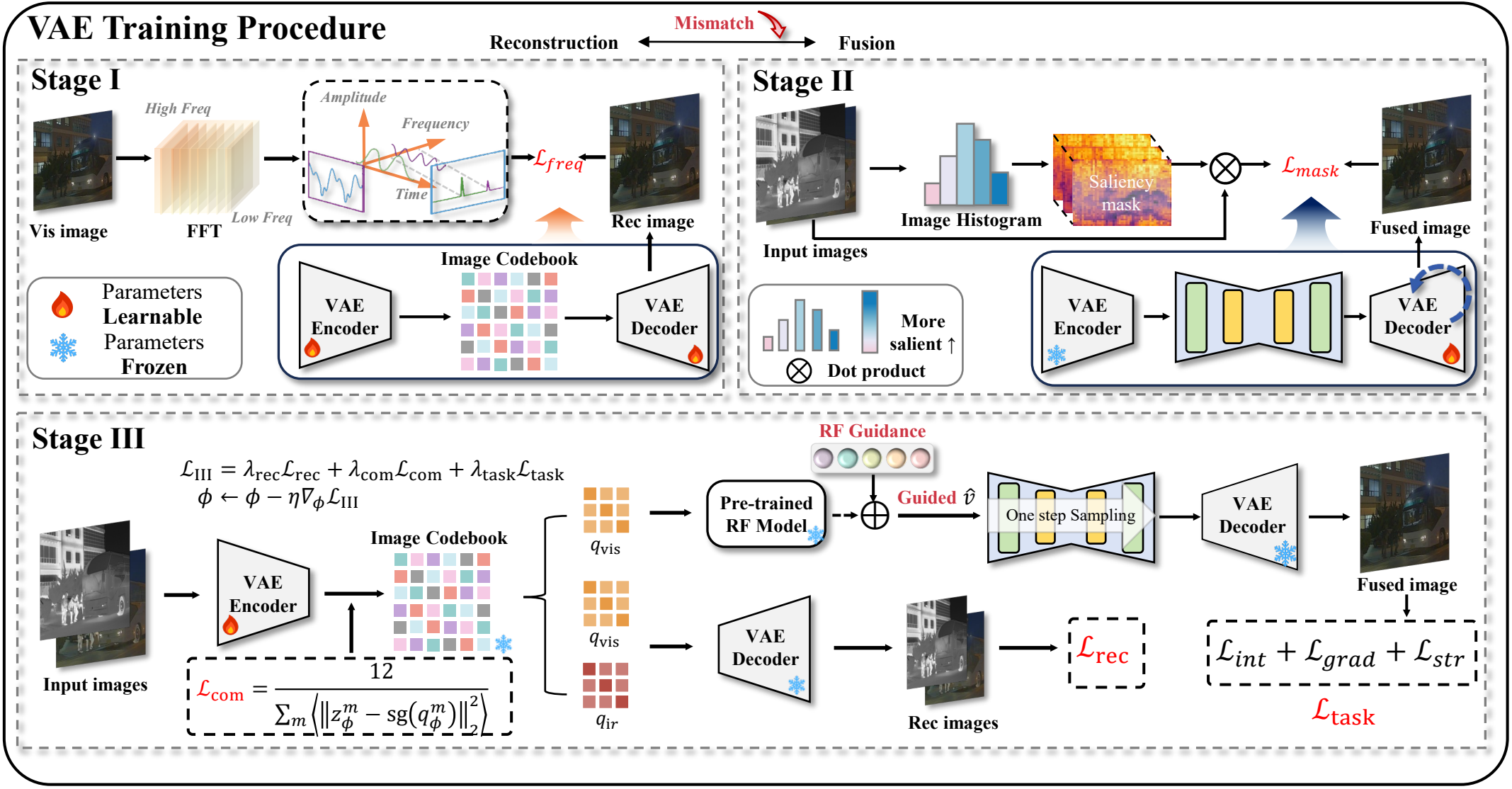

VAE Training Procedure
Reconstruction
Mismatch
Fusion
Stage I
High Freq
Low Freq
Amplitude
Frequency
Time
Vis image
FFT
$\mathcal{L}_{freq}$
Rec image
Image Codebook
VAE Encoder
VAE Decoder
Parameters Learnable
Parameters Frozen
Stage II
Input images
Image Histogram
Saliency mask
$\mathcal{L}_{mask}$
Fused image
More salient ↑
Dot product
Stage III
$\mathcal{L}_{\text{III}} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{com}}\mathcal{L}_{\text{com}} + \lambda_{\text{task}}\mathcal{L}_{\text{task}}$
$\phi \leftarrow \phi - \eta\nabla_{\phi}\mathcal{L}_{\text{III}}$
RF Guidance
Pre-trained RF Model
Guided $\hat{v}$
One step Sampling
q_{vis}
q_{vis}
q_{ir}
Input images
$\mathcal{L}_{\text{com}} = \frac{12}{\sum_m \left\langle \left\| z_\phi^m - \text{sg}(q_\phi^m) \right\|_2^2 \right\rangle}$
Rec images
$\mathcal{L}_{\text{rec}}$
$\mathcal{L}_{int} + \mathcal{L}_{grad} + \mathcal{L}_{str}$
$\mathcal{L}_{\text{task}}$